

MAC0623 - 3D Interaction in Mixed Realities

Assignment 1: Desktop Docking Testbed

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Public URL: <https://leonardomurakami.github.io/mac0623-a1/>

Repository: <https://github.com/leonardomurakami/mac0623-a1>

Browsers tested: Firefox and Chrome on Windows

Abstract

I compared two mouse mappings for a 6-DoF docking task, where the user must match a cube's position and orientation to a target.

The baseline switches between translation and rotation modes, while my mapping uses a 3D gizmo with separate handles for each axis. I expected the gizmo to improve accuracy at a small cost in speed.

With 5 participants and 31 recorded trials, the gizmo produced substantially lower position and orientation errors. Completion time was similar overall, although the effect on speed varied considerably between participants.

1 Method

The experiment was implemented in Three.js and run in Firefox and Chrome on Windows. The target tolerance was 0.05 scene units for position and 10° for orientation.

The two mappings were:

- **Baseline:** Spacebar or Tab switches between translation and rotation. Mouse movement controls two axes and the mouse wheel controls the third.
- **Gizmo:** Spacebar or Tab switches between translation and rotation handles. The user selects an axis handle and drags it, changing only that axis.

I tested 5 participants recruited from friends. Three were Computer Science students, one was a medical student, and one was from outside the CS group. P01 and P02 used the gizmo first, while P03, P04, and P05 used the baseline first.

Each participant received one untimed practice trial per mapping followed by three timed trials. One additional baseline trial from P03 was recorded, so the dataset contains 31 trials in total. No coaching was given during timed trials.

For each trial I recorded completion time, final position error, final orientation error, mode switches, and path length.

2 Results

Because participants completed both mappings, I also compared each participant's mean performance. On average, the gizmo was 2.16 s slower, reduced position error by 0.165 units, and reduced orientation error

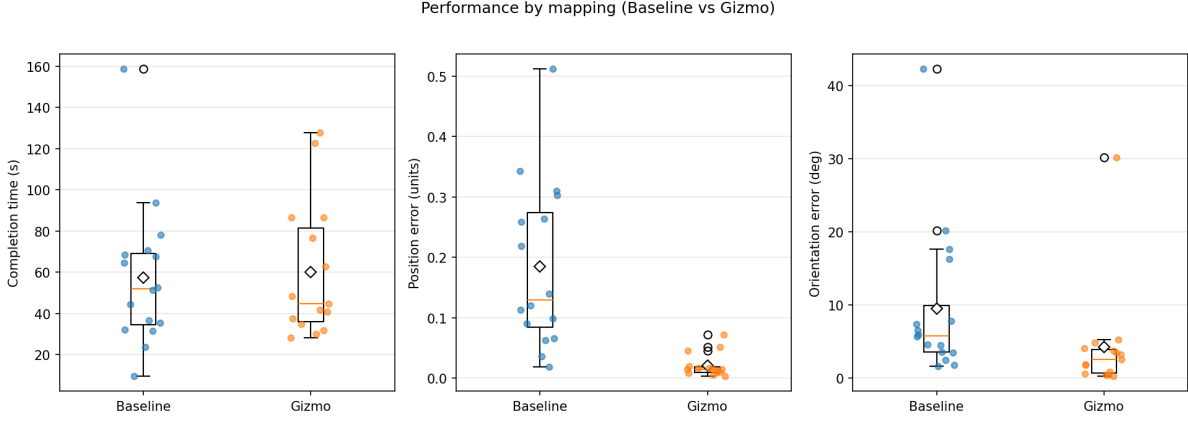


Figure 1: Performance by mapping.

Metric	Baseline	Gizmo	n
Completion time	57.45 ± 34.95 s	60.04 ± 32.87 s	16 / 15
Position error	0.185 ± 0.137	0.021 ± 0.019	16 / 15
Orientation error	$9.49^\circ \pm 10.45^\circ$	$4.23^\circ \pm 7.37^\circ$	16 / 15
Mode switches	5.75 ± 4.65	4.20 ± 2.60	16 / 15

Table 1: Mean $\pm$ standard deviation. The final column shows Baseline / Gizmo trial counts.

by 5.72° .

Position accuracy was the clearest result. Every participant had lower position error with the gizmo, and the overall mean fell from 0.185 to 0.021, an improvement of about 8.8 times.

Orientation accuracy also improved for every participant except P02. Overall mean orientation error fell from 9.49° to 4.23° .

Speed was less consistent. P01, P02, and P03 were slower with the gizmo, while P04 and P05 were faster. The largest differences were P03, who was about 58 s slower, and P04, who was about 67 s faster.

3 Discussion

The gizmo clearly performed better on **accuracy**, while neither mapping clearly won on **speed**. This partially supports my original prediction: the expected accuracy improvement appeared strongly in the data, but the speed cost was small overall and depended heavily on the participant.

The position result fits the motivation for the design. The baseline has a **DoF mismatch**: a 2D mouse controls a 6-DoF task by decomposing it into translation and rotation modes. The gizmo decomposes it further, with one axis controlled per drag. This prevents a correction on one axis from changing another, which likely explains the much smaller position errors.

Orientation showed a similar but weaker effect. Four of the five participants improved with the gizmo. P02 was the exception, with mean orientation error increasing from 8.48° to 11.31° . This may indicate a cost in learning or selecting the rotation handles: constraining the movement helps only after the user understands which ring controls each axis.

The speed results were more surprising. P03 was much slower with the gizmo, suggesting that selecting individual handles added substantial overhead. P04 showed the opposite behavior, dropping from a mean

of 101.7 s with the baseline to 34.9 s with the gizmo. For this participant, avoiding repeated corrections appears to have saved more time than handle selection cost.

There are several limitations. The sample is still small and was recruited from my own friends rather than randomly. Three participants were CS students, so familiarity with interfaces similar to 3D editing tools may also affect the results. The mapping order was counterbalanced as evenly as possible with five participants, but practice effects may still remain. Finally, one additional baseline trial was recorded for P03, producing slightly unequal trial counts between mappings.

All of the participants said that the Gizmo was more intuitive to learn and control the cube while the Baseline was extremely confusing and frustrating because it felt unnatural.